Sukhraj Purewal

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Education

Simon Fraser University Burnaby, B.C.
Bachelor of Science in Computer Science *May 2027*
Relevant Coursework: Software Engineering, Intelligent Systems, Operating Systems, Data Structures and Algorithms, Artificial Intelligence, Database Systems, Linear Algebra, Web-Development, and Statistics.

Work Experience

Software Developer Intern Jan 2025 to Aug 2025
Astera Labs *Vancouver, BC*

- Built a **Python** RAG tool over internal docs, improving issue triage and onboarding by **30%**.
- Implemented a **C++** API layer on the networking SDK for configuring and monitoring devices in large-scale data centers.
- Created a test suite in **Python** and **Bash** for deepening device logging across memory and network systems, cutting troubleshooting time by **35%** with fully traceable logs.
- Developed a containerized benchmarking toolbox for memory and networking devices with automated parsing, trend analysis, and live dashboards, improving developer efficiency by **15%**.
- Collaborated with firmware, platform, and data teams for resolving memory-tiering issues and standardizing reproducible debug workflows for enterprise customers (Meta, Microsoft, AWS).

Software Engineer Intern Sep 2023 to May 2024
MineSense Technologies Ltd. *Vancouver, BC*

- Automated **Jenkins** CI/CD pipelines on an on-prem **Kubernetes** cluster, cutting deploy time by **30%**.
- Added **Regex**-based configuration validation, catching **20%** more integration bugs and reducing diagnostic time by **15%**.
- Expanded **Selenium**/**pytest** coverage with automated reports and flaky-test triage, reducing bug volume by **30%**.
- Partnered with data engineers to align validation datasets with field conditions, increasing model fidelity by **13%**.

Extracurricular Experience

Software Development Team Lead May 2024 to Present
SFU Robotics Soccer Club *Burnaby, BC*

- Lead a team of more than twenty developers building networking infrastructure, GUI simulations, and AI agents in **C++**.
- Implemented a D Star Lite and Conflict Based Search path planning system for coordinated multi robot navigation.
- Built a physics engine for accurate motion simulation that reduced real world debugging time by **30%**.
- Designed real time network control systems for fast communication and responsiveness during matches.

Projects

PathWars Aug 2025

- Designed and built a real time strategy game in **C++** and **Qt** where players control autonomous robots that compete for goals on a dynamic grid environment.
- Implemented **D Star Lite** and **Conflict Based Search** algorithms for multi agent path planning and obstacle avoidance.
- Developed a game physics engine for motion simulation, collision response, and agent communication.
- Added a **WebSocket** server for multiplayer support and spectator visualization with live scoring and replay playback.

Auto Car July 2025

- Built a self driving vehicle on a **Raspberry Pi 4** with **Pi Camera V2** and custom motor drivers, handling perception, control, and telemetry.
- Integrated vision based SLAM and A* navigation with PID motion control to maintain less than **10%** of path deviation.
- Created a **Flask** dashboard for live video streaming, parameter tuning, and OTA updates, cutting debugging time by **30%**.

NBA Championship Predictor June 2025

- Developed a machine learning model using two decades of NBA data including team records, injuries, and performance metrics to predict playoff outcomes.
- Automated data collection and nightly updates using **Python** and **pandas** pipelines that scrape league statistics.

Skills

Languages: Python, JavaScript, C++, Go, C, Java, CSS, HTML, C#, Bash, SQL
Frameworks: Selenium, Pytorch, Spring, React, Flask, .NET, Qt, Django, Node.js
Developer Tools: AWS, VS Code, Jenkins, Docker, Git, MySQL, Kubernetes, MongoDB